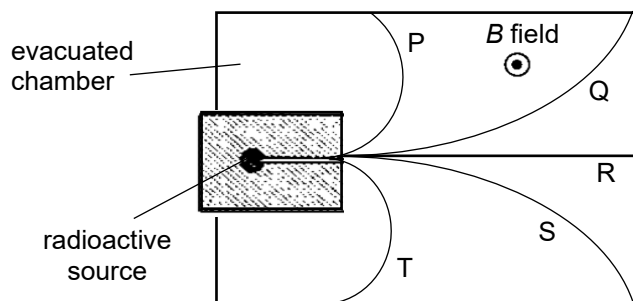


- 30 A source undergoing alpha, beta and gamma decay is placed in an evacuated chamber with magnetic field directed out of the page. Which of the following represents the paths of the radiation particles emitted?



	α -particle	β -particle	γ -ray
A	Q	T	R
B	S	P	R
C	T	R	S
D	S	T	Q

Ans: B

α -particle: Positively charged, so will deflect downwards in the B-field (use FLHR).

β -particle: Negatively charged, so will deflect upwards in the B-field.

From $Bqv = mv^2/r \rightarrow q/m$ is inversely proportional to r
 (q/m) of α -particle $< (q/m)$ of β -particle, so $r_\alpha > r_\beta$

γ -ray: Has no charge so it is not deflected

End of Paper